

On the behaviour of charged particles in Electromagnetic fields

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Abstract

In this experiment, values of two physical constants of high significance are found. A PASCO Model Coulomb Balance was used together with a kV power supply to obtain a value for the permittivity of free space. Using the experiment originally carried out by Coulomb, this value was found to be $(4.19 \pm 0.26) \cdot 10^{-12} Fm^{-1}$ which is around half of the literature value. This significant deviation will be discussed along with methods of obtaining more accurate results. The charge-to-mass ratio of the electron is also determined using a T-Model Fine-Beam tube. This was found to be $(1.56 \pm 0.031) \cdot 10^{11} Ckg^{-1}$, which is in relatively close agreement with current literature. The e/m ratio was found to be $(1.75 \pm 0.033) \cdot 10^{11} Ckg^{-1}$ when an electron beam with a 5cm diameter was used which is almost exactly the currently accepted value. In addition, the relationships between the motion of electrons and the strength of the **E** and **B** fields were also observed. It was found that the strength of the electric field was proportional to the kinetic energy of the electrons and the strength of the magnetic field was proportional to the curvature of the paths along which the electrons travelled.

1 Introduction

The primary aim of this experiment is to gain a deeper understanding of electromagnetic phenomena. Through the study of two of these effects, namely the Coulomb force and the behaviour of electrons in $\mathbf{E}$ and $\mathbf{B}$ fields, values of ϵ_0 and e/m_e are obtained. The experiment Coulomb carried out in 1785 is recreated although for a different purpose. Originally, he used the experiment to prove the inverse square law for electrostatics.[1]

$$F_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2} \quad (1)$$

where Q_1 and Q_2 are two point charges and r is the distance between them. Equation (1) is assumed to be true and instead, the goal is to obtain a value for the permittivity of free space. It is important to note that this law only applies to point charges, however, we can use two charged spheres (Figure 1) provided that they are a sufficient distance apart. This works as with a large distance, r , the charge distributions on each sphere will stay uniform and thus be equivalent to a point charge at the centres of the spheres.

Continuing the study of charged particles, we place electrons in an electromagnetic field. The method employed is similar to the experiment conducted by J.J Thomson in 1897.[2] We vary the strength of the $\mathbf{E}$ and $\mathbf{B}$ fields and extrapolate the electron's charge to mass ratio from the way electrons respond to these changes. This is done by the application of the Lorentz force.[3]

$$\mathbf{F}_L = m \frac{d^2 \mathbf{r}}{dt^2} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \quad (2)$$

where q is the charge of the electron, $\mathbf{v}$ is the velocity of the electron due to an electric field, $\mathbf{E}$ is the electric field and $\mathbf{B}$ is the magnetic field. In addition, the separate effects of the electric and magnetic fields on the path of the electron are observed. It's confirmed that the strength of the magnetic field is proportional to the curvature of the electrons' path and that the strength of the electric field is proportional to the kinetic energy of the electrons. Similarly, the inverse square law is verified for electrostatics by showing that Equation (1) produces straight line graphs when we take r at quadratically spaced points.

2 Method

In this experiment, two main pieces of apparatus were used: a PASCO Model Coulomb Balance and a T-Model Fine-Beam Tube with the Coulomb Balance used to measure ϵ_0 and the Fine-Beam Tube used to find the electron's charge-to-mass ratio. For the Coulomb Balance, a charging probe connected to a kV power supply was used to keep two conducting spheres charged. This was necessary as, once charged, both spheres would become electrically neutral after a short amount of time.

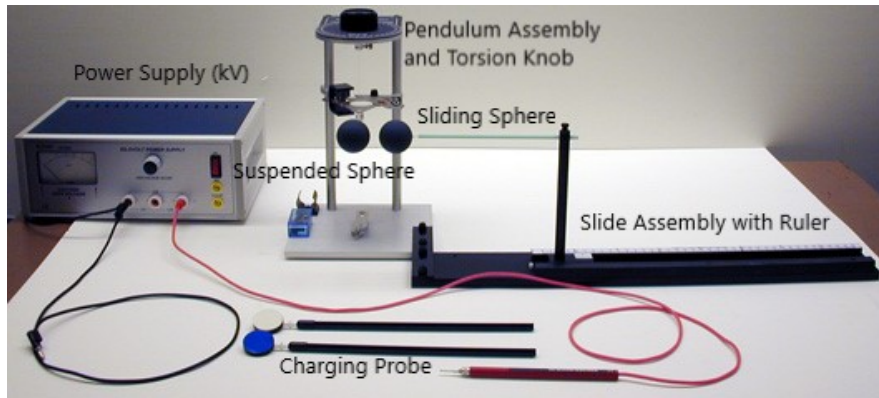


Figure 1: An experimental setup identical to the one used in the laboratory with a PASCO Model Coulomb Balance

We first measured the angle between the Sliding and Suspended spheres at different distances between them while keeping both spheres charged. Plotting a graph of θ against $1/r^2$, we obtained a straight line graph. The gradient of this graph is given by the following expression which we rearrange to find ε_0

$$m = k \frac{C_S^2 V_S^2}{K_{tor}} \implies \varepsilon_0 = \frac{m K_{tor}}{4\pi r V_S^2} \quad (3)$$

where m is the gradient, k is the proportionality constant from Equation (1), K_{tor} is the torsion constant and C_S and V_S are the capacitance of and voltage across the conducting spheres respectively. Of course, after this, we measured the torsion constant. This was done by separating the Coulomb Balance from the Slide Assembly and turning it on its side. A lateral support beam was placed under the Suspended Sphere. The sphere was then carefully balanced with small copper rings placed on the end of the pendulum opposite the sphere until it hovered just above the support beam. Masses of 20mg, 40mg, 50mg and 70mg were separately placed on the centre line of the sphere. With each mass, we once again balanced the pendulum by turning the Torsion knob so that the sphere hovered above the support beam. A graph of Torsion angle against mass was then plotted and K_{tor} calculated from its gradient.

Next, we changed to the Fine-Beam Tube. An electron gun inside a glass chamber is used to create an electron beam. The chamber is filled with dense Helium. This is to ensure that some fraction of electrons produced by the electron gun collide with and thus excite some of the Helium atoms, emitting energy in the form of visible light. We can then observe how this 'light beam' behaves when changes are made to the electromagnetic field created around the chamber.

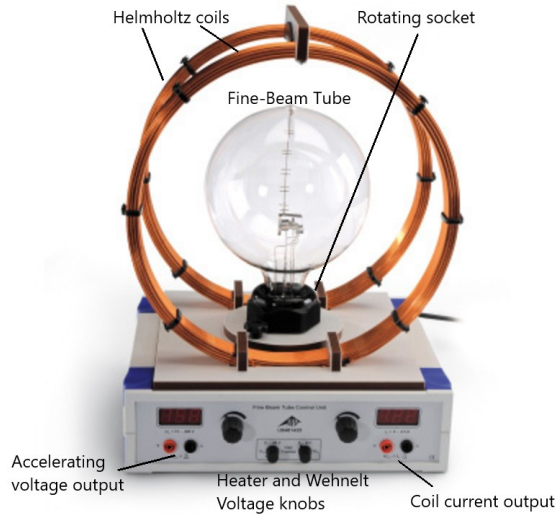


Figure 2: A T-Model Fine Beam tube used to measure the e/m_e ratio.

With the Voltage and Current outputs connected to a power supply, we began by varying the voltage at set intervals of 20V. With each new voltage value, the current was also adjusted so that the radius of the electron beam remained constant. Then, we varied the current at set intervals of 0.1A and adjusted the voltage so ensuring that the radius once again is unchanged. This process was repeated for radii of 2.5cm, 2.0cm and 1.5cm and at each r the values of current and voltage were recorded. The data collected was then used to produce graphs of U , the potential difference produced by the electric field, against I^2 , the square of the coil current (analogous to the strength of the $\mathbf{B}$ field). The electron's charge to mass ratio was then found using the following expression

$$\frac{e}{m_e} = \frac{2U}{\gamma^2 I^2 r^2} = \frac{2}{\gamma^2 r^2 g} \quad (4)$$

where γ is $7.43 \cdot 10^{-4} \text{kg s}^{-1} \text{A}^{-1}$, r is the radius of the electron beam and g is the gradient of the $U-I^2$ graph for a beam of radius r .

3 Experimental Results

Note: all gradient values and uncertainties in gradients were calculated using Excel's LINEST function unless stated otherwise.

A linear relationship was found, as predicted, between the torsion angle and $1/r^2$ where r is the distance between the spheres. The gradient of the $\theta-1/r^2$ graph as well as of the $\theta-m$ graph were used to calculate ε_0 using Equation (3).

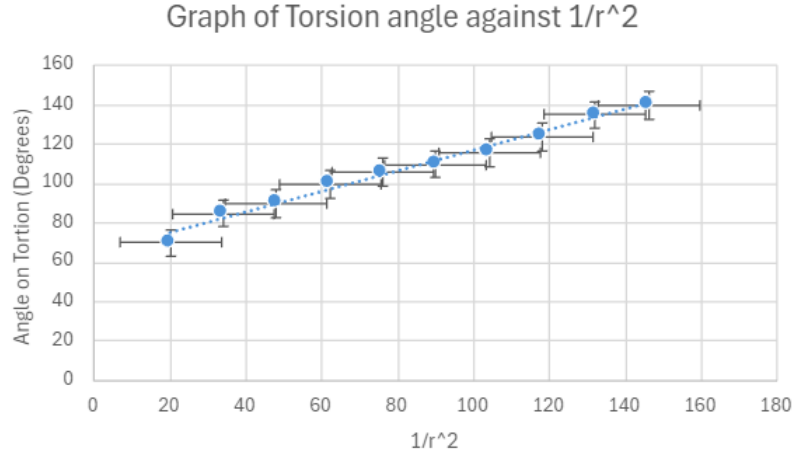


Figure 3: Graph of Torsion angle against reciprocal square of distance between spheres in cm^{-2} .

The observed value of ε_0 was $(4.19 \pm 0.26) \cdot 10^{-12} \text{Fm}^{-1}$ with the literature value being $8.85 \cdot 10^{-12} \text{Fm}^{-1}$.^[4] Similarly, it was found that the Voltage used by the Electric field in the Fine-Beam tube is proportional to the square of the current through the Helmholtz coils.

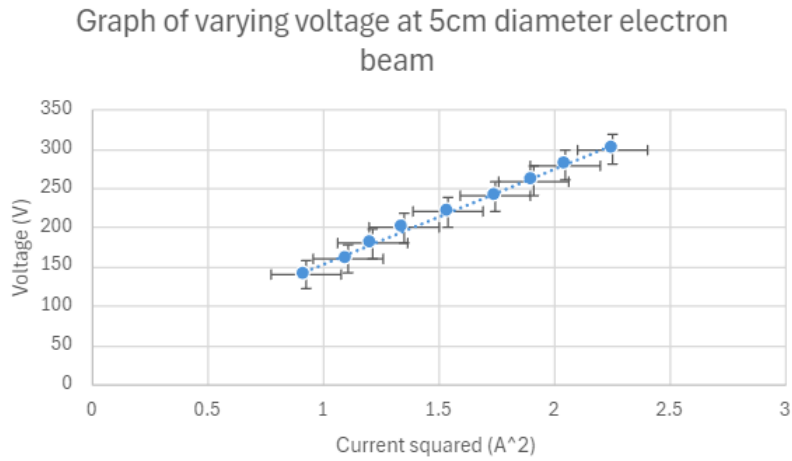


Figure 4: Graph of Varying voltage against square of current for a 5cm diameter electron beam. 6 such graphs were produced showing the results as voltage and current were separately varied for diameters of 5cm, 4cm and 3cm all showing the same relation as seen in this figure. An average of the gradients of these graphs was taken.

Hence, the electron's charge-to-mass ratio was found to be $(1.56 \pm 0.031) \cdot 10^{11} \text{Ckg}^{-1}$. The literature value for this is $1.76 \cdot 10^{11} \text{Ckg}^{-1}$.^[4]

4 Discussion

While not explicitly shown, it is observed that the kinetic energy of the electrons is proportional to the strength of the Electric field and the curvature of the electron beam is proportional to the strength of the Magnetic field. When varying either the voltage or the current (i.e.: $|\mathbf{E}|$ and $|\mathbf{B}|$ respectively), the radius of the electron beam increased as voltage increased and decreased as the current increased. However, when the magnetic field strength was 0Am^{-1} , the electron beam had no curvature regardless of the strength of the electric field. Hence, the magnetic field strength must be proportional to the curvature of the electron beam path. The change in curvature due to the change in the electric field can be explained by an increased kinetic energy: if the electrons' kinetic energy increased, then their velocity also increased, meaning they travelled further per unit time hence increasing the beam radius.

The values of ε_0 and e/m_e were found to have deviations of 52.3% and 11.4% from literature values respectively.[4] While the value found for the charge-to-mass ratio is in rough agreement with the literature value, there is a severe gap between the true permittivity of free space and the observed value. This could be due to many different factors, the most likely of which is the high uncertainty in the torsion angle.

A Coulomb Balance is a very delicate piece of equipment: it is quite difficult to measure the torsion angle precisely as the Suspended sphere never fully stops oscillating. Hence, there will always be a significant error in the measurement of θ if this is done by a human. One possible solution is to use a software to measure this angle, reducing most of the error that comes from the unreliability of human judgement. Another possible cause for this disagreement is any static charge picked up by clothing close to the spheres. It's best to dress in cotton clothes or, better yet, short sleeve T-shirts so as to avoid close contact with the spheres by anything other than the charging rod.

It is also worth mentioning that the accuracy of values for e/m_e decreased exponentially with each graph. The first set of data collected found this ratio to be $1.75 \cdot 10^{11}\text{Ckg}^{-1}$ which is almost exactly equal to the literature value.[4] Although this is only speculation, this may imply that the accuracy of this measurement gets better for higher values of the electron beam radius and thus asymptotically approaches 100% for increased electric field strength.

5 Conclusion

Values of both ε_0 and e/m_e were found with relatively low uncertainties of $(4.19 \pm 0.26) \cdot 10^{-12}\text{Fm}^{-1}$ and $(1.56 \pm 0.031) \cdot 10^{11}\text{Ckg}^{-1}$ respectively. The relation between the motion of electrons and the strength of the electric and magnetic field were observed using the Fine-Beam Tube. Similarly, the inverse square law for electrostatics was confirmed using a Coulomb Balance. Hence, all main goals of the study were achieved.

References

- [1] Coulomb, Charles Augustin, (1785) *Premier mémoire sur l'électricité et le magnétisme*. Histoire de l'Académie Royale des Sciences. Imprimerie Royale.
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- [4] W.H.J. Childs, (1972) *Physical Constants*, Science Paperbacks, 9th ed